# TempTrek

## Environmental Monitoring System — Installation Manual

A clear guide for installing, starting, and checking TempTrek on Windows.

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|---|---|
| **Application version** | 2.5.5 (Windows x64) |
| **Intended audience** | System administrators and authorized TempTrek users |
| **Supported package** | MSI installer or standalone executable |
| **Document purpose** | Installation, first-time setup, basic verification, and recovery guidance |

**Before you begin:** keep the installation package, activation key, Super Admin password, and backup password secure. They are separate credentials.

Prepared for TempTrek Environmental Monitoring System v2.5.5.

# 1. Before installation

Use a 64-bit Windows computer with network access to the Modbus TCP devices that will be monitored. The delivered application bundles its Python runtime; end users do not need to install Python.

| Requirement | Minimum | Recommended |
|---|---|---|
| Operating system | Windows 10 or later, 64-bit | Windows 11 or supported Windows Server, 64-bit |
| Memory | 2 GB RAM | 4 GB RAM or more |
| Storage | 250 MB for the application | Additional space for CSV readings, reports, and backups |
| Network | Access to Modbus TCP devices | Stable dedicated LAN where practical |
| Permissions | Windows user account | Administrator rights for installation and backup management |

> **Important:** TempTrek requests Windows elevation when it launches. Backup and restore management additionally requires an elevated Windows administrator session and the application user's *Backup & Restore* permission.

# 2. Choose an installation method

| Package | Use it when | File |
|---|---|---|
| MSI installer | You want a normal Windows installation with shortcuts and uninstall support. | **TempTrek-2.5.5-x64.msi** |
| Standalone executable | You need to run the program directly without installing it. | **TempTrek Environmental Monitoring System.exe** |

The packages are unsigned. Obtain them only from your approved TempTrek delivery source and verify the supplied SHA-256 checksum before production deployment.

### Install with the MSI (recommended)

1. Close any running TempTrek application.
2. Right-click **TempTrek-2.5.5-x64.msi** and choose **Install**, or double-click it.
3. Approve the Windows UAC prompt when it appears.
4. Complete the setup wizard. The default installation location is under Program Files.
5. Start TempTrek from the Start menu or desktop shortcut.

### Run the standalone executable

1. Copy the executable to an approved local folder. Do not run it from an email attachment or temporary download location.
2. Double-click the executable and approve the UAC prompt.
3. Keep the executable available for future launches; application data is stored separately in your Windows profile.

## 3. First launch and activation

1. Open TempTrek and approve the Windows UAC prompt.
2. If prompted, enter the supplied activation key. Activation validates the key for the current computer.
3. On a new Windows profile, create the **Super Admin** password. Use at least six characters and store it according to your site password procedure.
4. Sign in using the new Super Admin account.
5. Confirm that the dashboard opens without an error message.

> **Do not delete configuration files:** an existing installation with a missing or damaged user database intentionally fails closed. Restore a valid encrypted backup instead of trying to recreate accounts.

## 4. Configure the system

Complete these steps using an authorized account before placing the system into service.

1. Open **Administration** → **Device Manager**. Add each Modbus TCP device: IP address, port, slave ID, save interval, and register mappings.
2. Choose the matching Modbus function and data type for every channel: coils/discrete inputs or holding/input registers; Boolean, integer, float, BCD, and related supported formats.
3. Open **Notifications** to configure email or SMS alert rules as required. Test each notification route before relying on it.
4. Open **Administration** → **System Users** to create authorized user accounts and assign only the permissions each role needs.
5. Open **Administration** → **Company Details** to set the report header and optional logo.
6. Start monitoring from the dashboard. Confirm that values update and the timestamp is current.

> **Good practice:** configure one device first, verify its values against the physical instrument, then add the remaining devices.

## 5. Where TempTrek stores data

| Location | Contents |
|---|---|
| **%APPDATA%\TempTrek\config** | DPAPI-protected settings, user accounts, devices, approvals, and notification settings. |
| **%APPDATA%\TempTrek\live** | Daily CSV reading, audit, alarm, email, and SMS logs. |
| **%APPDATA%\TempTrek\reports** | Generated PDF reports and preserved restore-conflict material. |
| **%ProgramData%\TempTrekBackups** | Administrator-protected encrypted backup archives. |

Settings are encrypted for the Windows account that runs TempTrek. Moving configuration files to a different Windows account will not make them readable there; use an encrypted TempTrek backup and the backup password for transfer.

# 6. Back up and restore safely

## Create the first backup

1. Run TempTrek as a Windows administrator and sign in with an account that has **Backup & Restore** permission.
2. Open **Administration** → **Backup & Restore**.
3. Create and confirm a dedicated backup password (at least six characters). This password is not the same as the TempTrek login password.
4. Select **Create Backup Now** and wait for the success status.
5. Record the backup location and store the backup password separately and securely.

Backups use authenticated AES-GCM encryption. They include the protected configuration and the selected application data. Automatic backups can be configured from the same screen.

## Restore a backup

1. Stop operational work and ensure you have approval to restore.
2. Open **Administration** → **Backup & Restore** in an elevated administrator session.
3. Select the backup archive, authenticate when prompted, and enter the backup password.
4. Review the restore confirmation carefully. TempTrek creates a safety backup before it changes data.
5. After a successful restore, restart TempTrek when instructed. Monitoring is paused during the operation.

> **Never manually extract or overwrite TempTrek data folders during a restore.** The application validates the archive and uses a rollback transaction to protect the current data.

# 7. Installation check

Use this short check before declaring the installation ready:

1. TempTrek opens and the correct user can sign in.
2. Dashboard shows current values for a test device.
3. A test record appears in the expected daily live CSV file.
4. A test alert route succeeds, if notifications are enabled.
5. A report can be generated and is marked DRAFT or FINAL / APPROVED as appropriate.
6. A manual encrypted backup completes successfully.
7. The Audit Log records the configuration and test actions.

## 8. Quick troubleshooting

| Problem | What to check |
|---|---|
| Application will not start | Use the same Windows account that owns the existing settings. Do not replace configuration files. Restore a valid backup if the user database is missing or damaged. |
| Device is stale or disconnected | Confirm IP address, port, slave ID, Modbus function, network route, and device power. A failed read does not replace the last value with zero. |
| Backup controls are unavailable | Launch as a Windows administrator and confirm the signed-in TempTrek user has Backup & Restore permission. |
| Email test fails | Check the configured SMTP host, credentials, port, firewall, and server TLS certificate. TempTrek requires verified TLS. |
| Cannot open settings after a Windows account change | DPAPI protects settings for their original Windows account. Restore through TempTrek using a portable encrypted backup. |